

Cyber Security

Laporan Progress PJBL

Kelompok 1
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Konteks Project IoT

GeoSync merupakan modul traktor pertanian otonom canggih yang dilengkapi teknologi otomatisasi mekanis berbasis kendali jarak jauh (remote control) serta sistem kecerdasan buatan. Alat ini mengintegrasikan mekanisme pemutar engkol otomatis dan tuas dekompresi elektronik agar proses pembakaran awal (engine start) mesin diesel dapat dilakukan dari darat tanpa penanganan manual di sawah, dipadukan dengan sistem manajemen suhu otomatis (thermal management) untuk mencegah kemacetan akibat beban kerja ekstrem, serta modul machine learning untuk optimasi pemetaan dan penyesuaian lintasan pembajakan lahan secara mandiri.



Stack Teknologi Replika IoT Lab

PERANGKAT EDGE/NODE

ESP32-S3

Microcontroller & Sensor Client

- Firmware C++ Custom
- Protocol: MQTT via Wi-Fi
- MqttSecure Client Library

GATEWAY & BROKER

NVDIA Jetson Nano / Linux

IoT Edge Collector OS:

- Linux JetPack (Ubuntu-based 64-bit)
- Broker: Eclipse Mosquitto
- UFW Firewall & Fail2ban

MONITORING & SIEM

Wazuh/Suricata

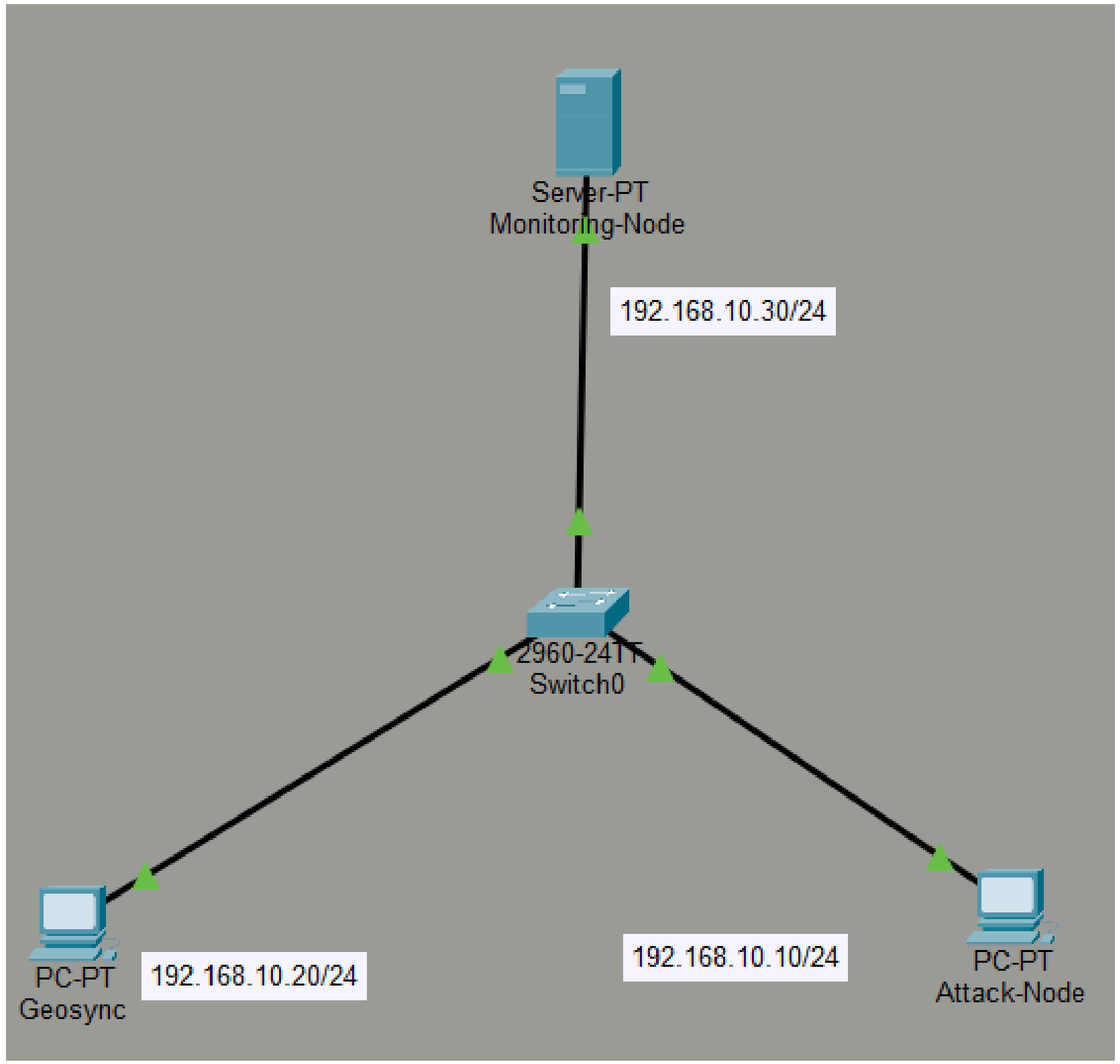
Blue Team Telemetry Center

- Suricata IDS Network Engine
- Sguil — Console analisis & korelasi alert real-time
- Centralized Log via tcpdump capture + pcap import



Topologi & Segmentasi Jaringan Virtual

Node	Role	IP Address	Sub net	OS
Attacker-Node	Red Team	192.168.10.10	/24	Kali Linux
Geosync	-	192.168.10.20	/24	Ubuntu Server
Monitoring Node	Blue Team	192.168.10.30	/24	Security Onion



BUKTI HARDENING

Kategori A

```
(venv) bagas@kelompok1:~/TEK1314-2026-Kel1-A1/backend-replika$ su - public
Password:
public@kelompok1:~$ touch /opt/shared-data/uji.txt
touch: cannot touch '/opt/shared-data/uji.txt': Permission denied
public@kelompok1:~$
```

Hardening Kebijakan Kata Sandi (Password Complexity Enforcement)

```
(venv) bagas@kelompok1:~/TEK1314-2026-Kel1-A1/backend-replika$ passwd bagas
Changing password for bagas.
Current password:
New password:
BAD PASSWORD: The password is the same as the old one
New password:
BAD PASSWORD: The password contains less than 1 uppercase letters
New password:
BAD PASSWORD: The password contains less than 1 uppercase letters
passwd: Have exhausted maximum number of retries for service
passwd: password unchanged
(venv) bagas@kelompok1:~/TEK1314-2026-Kel1-A1/backend-replika$
```

Hardening Hak Akses Direktori (Access Control & Least Privilege)

BUKTI HARDENING

Kategori B

Mengaktifkan firewall

```
bagas@kelompok1:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW IN    Anywhere
22/tcp (v6)                ALLOW IN    Anywhere (v6)

bagas@kelompok1:~$
```

Update patch sistem

```
   Main PID: 718 (unattended-upgr)
      Tasks: 2 (limit: 1011)
     Memory: 12.3M
        CPU: 190ms
     CGroup: /system.slice/unattended-upgrades.service
             └─718 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wa

Oct 22 14:18:49 kelompok1 systemd[1]: Started Unattended Upgrades Shutdown.
~
~
~
```

BUKTI HARDENING

Kategori B

Menutup port yang tidak digunakan

```
bagas@kelompok1:~$ sudo systemctl disable --now fwupd.service
bagas@kelompok1:~$ sudo systemctl disable --now ModemManager.service
Removed /etc/systemd/system/multi-user.target.wants/ModemManager.service.
Removed /etc/systemd/system/dbus-org.freedesktop.ModemManager1.service.
bagas@kelompok1:~$ sudo systemctl disable --now multipathd.service
Removed /etc/systemd/system/sysinit.target.wants/multipathd.service.
Removed /etc/systemd/system/sockets.target.wants/multipathd.socket.
Removed /etc/systemd/system/multipath-tools.service.
Warning: Stopping multipathd.service, but it can still be activated by:
  multipathd.socket
bagas@kelompok1:~$ sudo systemctl disable --now udisks2.service
Removed /etc/systemd/system/graphical.target.wants/udisks2.service.
bagas@kelompok1:~$ sudo systemctl disable --now upower.service
bagas@kelompok1:~$ systemctl list-units --type=service --state=running
  UNIT                          LOAD   ACTIVE SUB     DESCRIPTION
  cron.service                  loaded active running Regular background program processing daemon
  dbus.service                  loaded active running D-Bus System Message Bus
  getty@tty1.service            loaded active running Getty on tty1
  networkd-dispatcher.service   loaded active running Dispatcher daemon for systemd-networkd
  polkit.service                loaded active running Authorization Manager
  rsyslog.service               loaded active running System Logging Service
  snapd.service                 loaded active running Snap Daemon
  ssh.service                   loaded active running OpenBSD Secure Shell server
  systemd-journald.service      loaded active running Journal Service
  systemd-logind.service        loaded active running User Login Management
  systemd-networkd.service      loaded active running Network Configuration
  systemd-resolved.service      loaded active running Network Name Resolution
  systemd-timesyncd.service     loaded active running Network Time Synchronization
  systemd-udevd.service         loaded active running Rule-based Manager for Device Events and Files
  unattended-upgrades.service   loaded active running Unattended Upgrades Shutdown
  user@1000.service             loaded active running User Manager for UID 1000

LOAD   = Reflects whether the unit definition was properly loaded.
ACTIVE = The high-level unit activation state, i.e. generalization of SUB.
SUB    = The low-level unit activation state, values depend on unit type.
16 loaded units listed.
bagas@kelompok1:~$ _
```

BUKTI HARDENING

Kategori C

Memastikan kredential tidak hardcoded

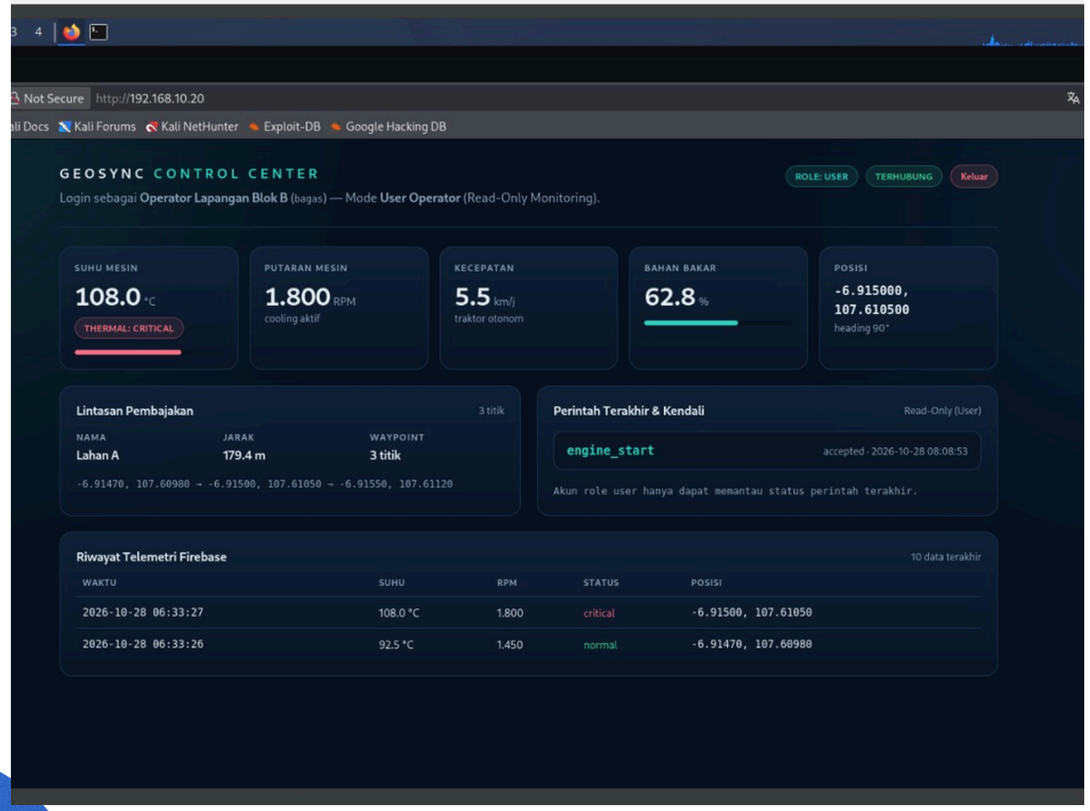
```
BASE_DIR = os.path.dirname(os.path.abspath(__file__))

load_dotenv(os.path.join(BASE_DIR, ".env"))
API_KEY = os.getenv("API_KEY")
SECRET_KEY = os.getenv("SECRET_KEY", "geosync-default-session-secret")
FIREBASE_DB_URL = os.getenv("FIREBASE_DB_URL")
FIREBASE_CRED_PATH = os.getenv("FIREBASE_CRED_PATH", "serviceAccountKey.json")
```

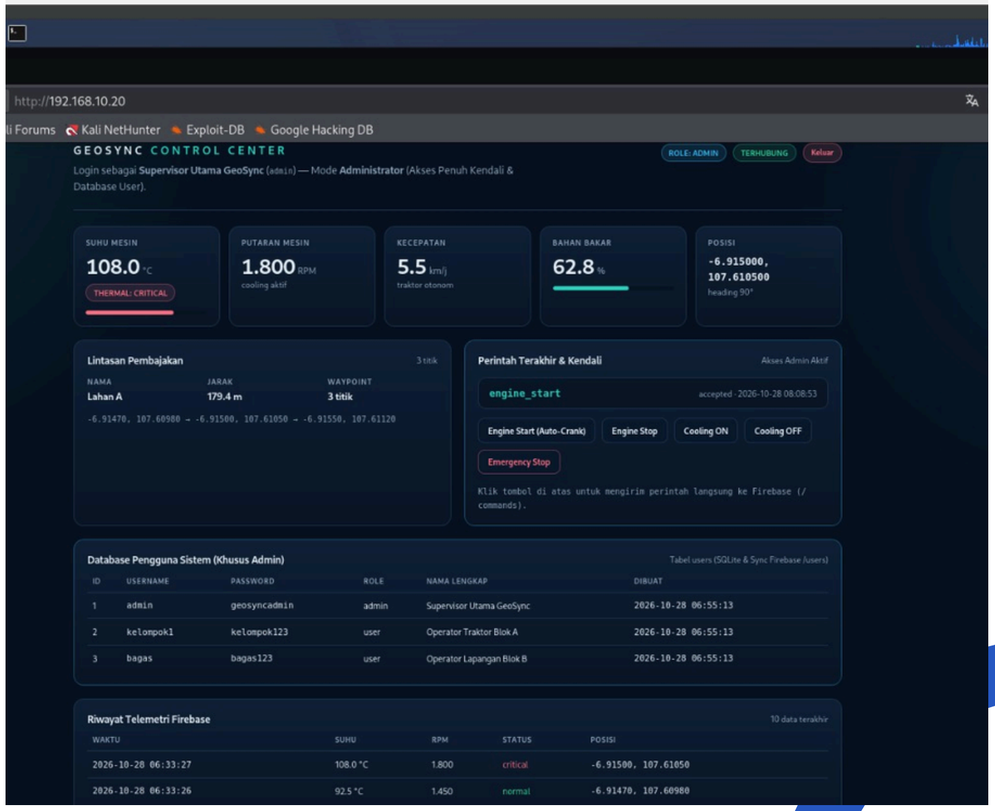
BUKTI HARDENING

Kategori C

Role base accsess control user



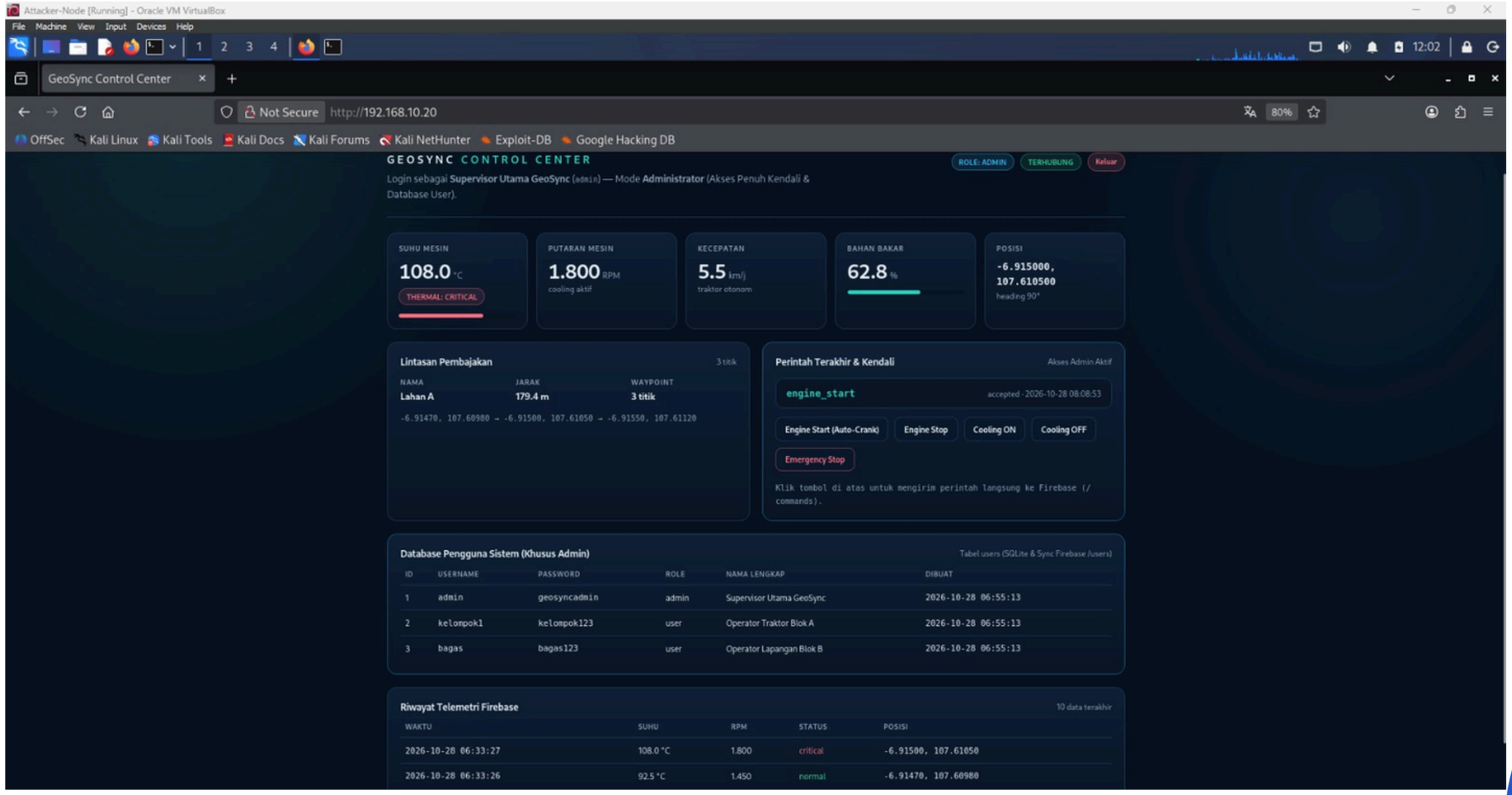
Role base accsess control admin



BUKTI HARDENING

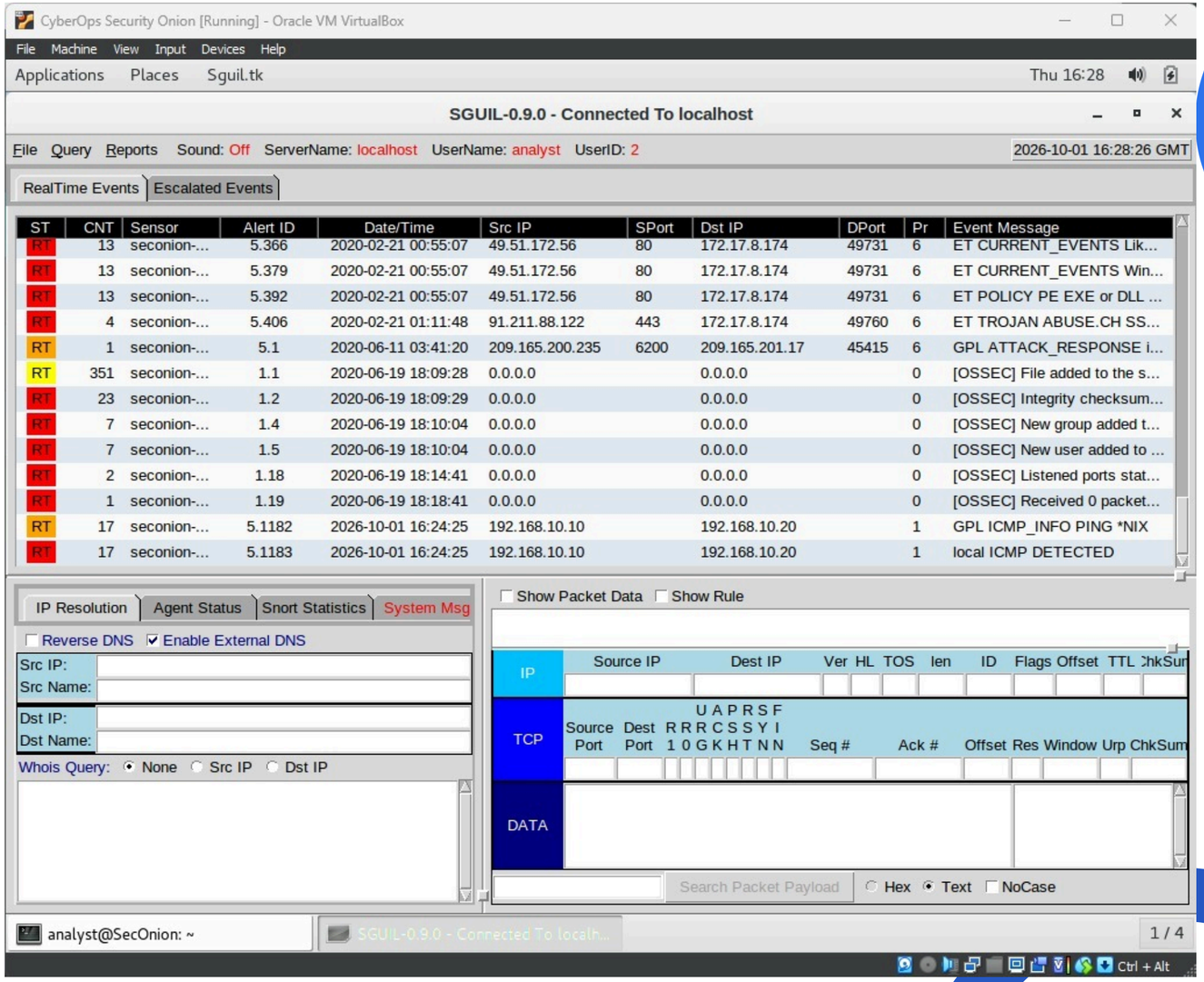
Kategori C

Role base accsess control



LIVE DEMO

Pengujian konektivitas dasar dilakukan dengan mengirim ICMP ping dari Attacker-Node (192.168.10.10) ke Target-Node (192.168.10.20). Traffic direkam menggunakan tcpdump pada interface monitoring eth0, kemudian diimpor ke Sguil untuk dianalisis.



Skenario Attack

1. Reconnaissance

Pemetaan layanan

Tools: Nmap

Uji: Port scanning, OS fingerprinting, service/version detection.

Target: Port 22 (SSH) pada Target-Node.

Deteksi Blue: Lonjakan koneksi SYN ke banyak port dari satu sumber IP.

2. Exploitation

Pengujian autentikasi

Tools: Hydra

Uji: Brute-force kredensial SSH (dictionary attack).

Target: Port 22, user admin.

Deteksi Blue: Login gagal berulang terdeteksi rule Snort, cross-check auth.log.

3. Availability

Ketersediaan & integritas jaringan

Tools: hping3, arpspoof

Uji: SYN flood ke SSH, ARP spoofing / MITM antar node.

Target: Port 22 & lalu lintas Layer 2 Target-Monitoring.

Deteksi Blue: Threshold SYN berlebih & alert ARP spoof di Sguil.



THANK YOU FOR YOUR ATTENTION

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https://github.com/cycwag/KELOMPOK1-KEAMANANSIBER

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